

A crash-course on boosting technical skills for writing scientific reports in $\text{\LaTeX}$

Part 2: Graphics, References & More

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Graphics

- ▶ Requires the `graphicx` package, which provides the `\includegraphics` command.
- ▶ Supported graphics formats include JPEG, PNG and PDF (usually).

```
\includegraphics[  
  width=0.5\textwidth]{gerbil}
```

```
\includegraphics[  
  width=0.3\textwidth,  
  angle=270]{gerbil}
```



Image license: CC0

Interlude: Optional Arguments

- ▶ We use square brackets `[ ]` for optional arguments, instead of braces `{ }`.
- ▶ `\includegraphics` accepts optional arguments that allow you to transform the image when it is included. For example, `width=0.3\textwidth` makes the image take up 30% of the width of the surrounding text (`\textwidth`).
- ▶ `\documentclass` accepts optional arguments, too. Example:
`\documentclass[12pt,twocolumn]{article}`
makes the text bigger (12pt) and puts it into two columns.
- ▶ Where do you find out about these? See the slides at the end of this presentation for links to more information.

Floats

- ▶ Allow $\text{\LaTeX}$ to decide where the figure will go (it can “float”).
- ▶ You can also give the figure a caption, which can be referenced with `\ref`.

```
\documentclass{article}
\usepackage{graphicx}
\begin{document}

Figure \ref{fig:gerbil} shows \ldots

\begin{figure}
\centering
\includegraphics[%
  width=0.5\textwidth]{gerbil}
\caption{\label{fig:gerbil}Aww\ldots.}
\end{figure}

\end{document}
```



Figure 1: Aww....

Figure 1 shows ...

- Put your references in a .bib file in 'bibtex' database format:

```
@Article{Jacobson1999Towards,
  author = {Van Jacobson},
  title = {Towards the Analysis of Massive Multiplayer Online
           Role-Playing Games},
  journal = {Journal of Ubiquitous Information},
  Month = jun,
  Year = 1999,
  Volume = 6,
  Pages = {75--83}}

@InProceedings{Brooks1997Methodology,
  author = {Fredrick P. Brooks and John Kubiawicz and
           Christos Papadimitriou},
  title = {A Methodology for the Study of the
           Location-Identity Split},
  booktitle = {Proceedings of OOPSLA},
  Month = jun,
  Year = 1997}
```

- Most reference managers can export to bibtex format.

bibT_EX 2

- ▶ Each entry in the .bib file has a *key* that you can use to reference it in the document. For example, Jacobson1999Towards is the key for this article:

```
@Article{Jacobson1999Towards,  
  author = {Van Jacobson},  
  ...  
}
```

- ▶ It's a good idea to use a key based on the name, year and title.
- ▶ L^AT_EX can automatically format your in-text citations and generate a list of references; it knows most standard styles, and you can design your own.
- ▶ **Extract** a citation from a random paper

bibT_EX 3

- ▶ Use the natbib package¹ with `\citet` and `\citep`.
- ▶ Reference `\bibliography` at the end, and specify a `\bibliographystyle`.

```
\documentclass{article}
\usepackage{natbib}
\begin{document}

\citet{Brooks1997Methodology}
show that \ldots. Clearly,
all odd numbers are prime
\citep{Jacobson1999Towards}.

\bibliography{bib-example}
% if `bib-example' is the name of
% your bib file

\bibliographystyle{plainnat}
% try changing to abbrnat

\end{document}
```

Brooks et al. [1997] show that Clearly, all odd numbers are pri
[Jacobson, 1999].

References

Fredrick P. Brooks, John Kubiawicz, and Christos Papadimitriou. A meth
ology for the study of the location-identity split. In *Proceedings of OOPSL*
June 1997.

Van Jacobson. Towards the analysis of massive multiplayer online role-playi
games. *Journal of Ubiquitous Information*, 6:75-83, June 1999.

¹There is a new package with more features named biblatex but most of
the articles templates still use natbib.

Exercise: Putting it All Together

Add an image and a bibliography to the paper from the previous exercise.

1. Download these example files to your computer.

[Click to download example image](#)

[Click to download example bib file](#)

2. Upload them to Overleaf (use the project menu).
3. Make use of the file *bib-example.tex*

More of TikZ

- ▶ $\text{\LaTeX}$ offers a variety of possibilities for drawing/plotting
- ▶ For detailed description visit <https://tikz.net/>
- ▶ For applications in electronics, take a look at *circuits_TikZ_1,2 and 3d.tex*

Rendering plots in L^AT_EX

- ▶ Plotting graphs directly in latex via *pgfplots* package
- ▶ Compile *results-plots.tex*
- ▶ Discuss the data preparation with headers
- ▶ For additional plots, refer to *MPC_PID* paper

Figures

- ▶ Figures should be prepared in vector format (WMF, EMF, PDF, ...) for good quality and small size
- ▶ Figures should not be prepared in bitmap format (JPG, BMP, PNG, GIF, TIF, ...)
- ▶ Figures can be prepared in vector format e.g. with
 - ▶ MATLAB (use Edit → Copy Figure and Paste → Paste Special → Picture (Enhanced Metafile))
 - ▶ MATLAB (use Edit → Copy Options → Figure Copy Template/Copy Options for configuration)
 - ▶ Simulink (use Edit → Copy Current View to Clipboard → Metafile and Paste → Paste Special ...)
- ▶ Remark:
 - ▶ Figures in vector format will not become pixelated when zooming
 - ▶ Figures in bitmap format will become pixelated when zooming

More of Figures

- ▶ Fonts

- ▶ The font in the figures should be the same as the font in the paper
- ▶ The font size in the figures should be minimally 6 pt

- ▶ Colors

- ▶ Colors should be used and chosen moderately
- ▶ Use printer colors

- ▶ Inkscape